National Technical University of Athens
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
DATABASE AND KNOWLEDGE SYSTEMS LAB

Databases

Spring Semester 2026

Course Description – Grading

Course Instructors

- Instructor (section1): Assoc. Prof. Dimitrios Tsoumakos
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- Instructor (section2): Dr. Vasileios Stefanidis
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- Laboratory Instructor/Coordinator: Dr. Marios Koniaris
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  - Laboratory Assistants: PhD students/graduate students of ECE

Info and how to contact us

Teaching:

- Theory: Tuesday 15:00-18:00, Amph. 2, room 007
- Lab: Monday 12:45-14:30, Wednesday 8:45-10:30

Course page: https://helios.ntua.gr/course/view.php?id=861

- Organization, announcements, material, submission of papers, etc.
- Slides-notes in English

- Ways of Communication: **(***very important***)**
  - Email: prerequisite for reply:
  - **From an INSTITUTIONAL ACCOUNT only**
  - **Clearly state your name and class**
  - **Report the course in the subject line**
  - Helios: use of forums

Course Learning Outcomes

- Knowledge of theoretical and technical topics for data management
- Knowledge of SQL
- (Installation) and operation of a Relational Database Management System (RDBMS)
- Design and management of a Relational Database
- Targeted practice on basic and advanced operational elements of a RDBMS

Course Syllabus

1. Entity Relationship model
2. Relational model
3. Relational algebra
4. The SQL query language
5. Physical organisation and storage
6. Hashing and indexing
7. Query Processing and Optimization
8. (possibly) Vectors – embeddings

Bibliography

- **Database System Concepts**, 7th Edition
  - https://www.db-book.com/
- **Database Management Systems**
  - Ramakrishnan & Gehrke
- **A First Course in Database Systems**, 3nd Edition
  - Jeff Ullman, and Jennifer Widom.
- **Database Systems: The Complete Book**
  - https://www.researchgate.net/publication/200034291_Database_Systems_The_Complete_Book

Grading

- Grading Scheme:
  - 30% Team Term Project (Mandatory)
  - 70% Final Exam (written, end of semester)

- Requirements for success in the course:
- (i) passing grade in the final exam
- (ii) Submission of the term project and passing grade on it
- written report + Oral exam + demo required for term project
- Specific exam schedule per group
  - **The project can be submitted ONLY in Spring**
  - **Term project grades (if >=50%) are permanently maintained**
  - **Grading stipulations (ρήτρες) do not apply**

Term Project

- Groups of <=3 students
- Design and implementation of a Relational Database with specific questions/functions for it. Very specific instructions will be given.

1. ER model
2. relational model
3. Implementing the database
4. queries/triggers/views, etc.
5. Optimization questions

Load/Requirements

- One of the most important courses in computer science
  - Basic principles of data management – prerequisites for every "data scientist“

- A lot of “theoretical” material will be covered
- The project offers practical coverage of basic theory chapters
  - Try to keep in touch with the theory
  - Try to work on the project during the semester and in the lab

- Start getting serious about engineering/devops aspects
- Installation-operation-debugging of systems, not just “using a given environment”

DATABASE – what is it?

- Organized collection of inter-related data that models some aspect of the real-world.
- Databases are the core component of most computer applications.
- Example:
  - Create a database that models a digital music store to keep track of artists and albums.
  - Things we need for our store:

    → Information about Artists

    → What Albums those Artists released